



Voraxia: Rebirth Foundation Game Design Document

**Multiplayer-first space engineering sandbox
foundation**

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Coding Custard Studios | Voraxia Documentation Standard



DOCUMENT PURPOSE

Foundation GDD

This document captures the current design direction for Voraxia: Rebirth. It establishes the game's identity, player journey, world structure, key systems, prototype evidence, and early-scope discipline.

CONFIRMED DIRECTION

PROPOSED FOR REVIEW

SCOPE / LATER LAYER

Working-document rule: "Proposed" entries are intentionally editable. They fill useful gaps without pretending those choices are already final.

Contents

[01 Project Identity](#)[02 High Concept & Player Fantasy](#)[03 Design Pillars](#)[04 Voraxia Prime & Spacegates](#)[05 Character Origins & Corporation](#)[06 New Player Experience](#)[07 Core Gameplay Loops](#)[08 Survival, Oxygen & Inventory](#)[09 Exchanges, Economy & Resources](#)[10 Raptor Mining & Asteroids](#)[11 Bases, Ships & Ship Editor](#)[12 Planetary Exploration](#)[13 Combat & PvP](#)[14 Multiplayer-First Principles](#)[15 Prototype Evidence & Vertical Slice](#)[16 Scope Discipline & Open Questions](#)

01 • PROJECT IDENTITY

Modern, embodied space engineering

Voraxia sits between the systemic seriousness of **Stationeers**, the construction scale of **Space Engineers**, and the embodied ship-and-station fantasy of **Starbase**, while aiming for a more coherent, high-fidelity, visually grounded experience.

Genre

Multiplayer-first space engineering sandbox with survival, construction, exploration, economy, PvP, ship design, and physically editable asteroid mining.

Core promise

Players mine, build, operate, repair, explore, trade, fight, and defend in a persistent industrial frontier where physical consequence matters.

Visual direction

Grounded industrial science fiction: believable metals, glass, machinery, cables, airlocks, dust, wear, practical lighting, and ships that feel constructed rather than toy-like.

Theme

Rebirth through reclamation. Raw asteroids, wreckage, damaged infrastructure, and broken frontier territory become new industry, new ships, new settlements, and new power.

Working tagline: Voraxia: Rebirth is a multiplayer space engineering sandbox about reclaiming a dangerous frontier, building a life within it, and defending what you create.

02 · HIGH CONCEPT & PLAYER FANTASY

A person inside the machine

Voraxia is a multiplayer space engineering sandbox in which players leave the protection of major stations, gather and trade resources, build modular bases and ships, travel through Spacegates into deeper regions of the Voraxia Prime star system, and survive a frontier where safety must be built, maintained, and defended.

“We made this. We run this. We can lose this.”

DESIRED PLAYER FEELING

What players should be able to do

Inhabit

Walk through a ship or station they helped create. Inspect machinery, enter airlocks, load cargo, and feel that every structure is a real place.

Reshape

Mine real asteroid material, salvage wreckage, repair damage, and leave visible consequences in the world.

Build

Create modular bases, industrial sites, ships, defences, logistics networks, and player-run infrastructure.

Risk

Travel beyond protected stations, face hostile space and other players, and understand that preparation, allies, and engineering change survival odds.

03 • DESIGN PILLARS

Rules that keep the game on course

1 Embodied engineering

Ships are places. Bases are places. Machinery is placed in the world. The player should stand inside the things they own and operate them as physical objects, not merely through remote menus.

2 Accessible depth

Voraxia should offer meaningful engineering without requiring Stationeers-level simulation density before players can enjoy themselves. Modular construction must be understandable; advanced construction should deepen possibility rather than become mandatory homework.

3 Earned safety

Major stations provide civilisation. Beyond them, safety is created through preparation, equipment, construction, placement, alliances, vigilance, and strength.

4 Physical consequence

Mining, construction, combat, storage, logistics, and damage should feel connected to objects and places. A ship carries cargo because cargo was loaded. A damaged ship needs repair because its systems matter.

5 Multiplayer-first world

Every major system must be designed for shared play, persistence, authority, fairness, performance, recovery, and exploit prevention from the beginning.

04 · WORLD STRUCTURE

Voraxia Prime

Voraxia Prime is a frontier star system containing protected stations, planets, moons, asteroid fields, asteroid belts, salvage sites, corporate infrastructure, player-built industry, and deeper regions unlocked through Spacegates.

Major stations

Stations are the anchors of civilisation: onboarding sites, social hubs, exchanges, repair points, resupply points, corporate facilities, travel nodes, and the primary protected environments.

Frontier space

Outside major station protection, there are no permanent safe zones. Mining, exploration, construction, trade, piracy, PvP, environmental hazards, and player-driven risk become real.

Spacegates

Spacegates are progression milestones, not merely loading doors. They allow travel deeper into Voraxia Prime, where opportunity, logistical difficulty, risk, and player conflict increase together.

Deeper regions may offer: rarer materials, large belts, salvage, strategic construction sites, difficult environments, contested routes, and increased PvP risk.

World rule: The station keeps players alive. The frontier makes them decide what that life is worth.

05 • CHARACTER ORIGINS

Two opening bargains

At character creation, players choose an origin path. This is not a permanent class prison. It establishes early resources, obligations, protection, commercial position, and social identity. Players may later change allegiance, work independently, join player organisations, or build their own industrial network.

Independent Miner

Freedom and vulnerability. Independent Miners own their direction, retain greater autonomy over their labour and goods, and choose their own contracts, routes, allies, and construction locations.

Benefits: no mandatory quotas; greater independence; stronger self-made identity.

Costs: no fleet protection; worse corporate exchange rates; less recovery, repair, logistics, and security support; higher exposure to piracy and frontier risk.

Corporate Astronaut

Security and obligation. Corporate Astronauts begin under contract to a powerful and unpleasant corporation. It provides equipment, work, fleet activity, logistics, better exchange rates, and limited protection.

Benefits: starter equipment; guided contracts; better rates; designated fleet protection; stronger repair and recovery access.

Costs: quotas; restricted choices; corporation takes a cut; reputation penalties for failure, desertion, theft, or disobedience.

“The corporation keeps you alive, but it does not keep you free.
Independence keeps you free, but space does not keep you safe.”

ORIGIN-PATH FRAMING

One corporation at launch

PROPOSED A single major corporation is the right launch-scope choice. It can influence contracts, exchanges, logistics, Spacegate access, security, starter equipment, and regional development without requiring a full constellation of competing brands and systems.

Corporate tone: It should be credible, useful, exploitative, and difficult to avoid. Players should think: “I dislike them, but I understand why people work for them.”

06 · NEW PLAYER EXPERIENCE

Safe enough to learn. Not safe forever.

Starting station

All players begin at a protected major station. It functions as the tutorial environment, social hub, exchange, supply point, corporate recruitment location, repair and recovery location, and first Spacegate access point.

Tutorial philosophy

For new players

Clear guided tasks teach movement, camera controls, interaction, mining, construction, ship operation, oxygen, inventory, and danger.

For experienced players

Accelerated onboarding, skippable guidance, optional deeper explanations, and contextual help that can be revisited later.

First departure

Leaving the station should be memorable. The player moves from structured safety into a world where resources matter, other players may be allies or threats, and a foothold must be earned.

Protected station

- > onboarding and preparation
- > first departure into unsafe space
- > gather and transport material
- > establish a foothold
- > build, engineer, defend, and expand

07 · CORE GAMEPLAY LOOPS

From resource run to frontier life

Primary loop

Prepare at station / base / ship

- > travel to an opportunity
- > mine, salvage, trade, fight, build, repair, or negotiate
- > move resources into cargo
- > process, sell, store, fabricate, or use them
- > improve capability and access
- > travel further into Voraxia Prime

Early-game loop

1. Complete or accelerate through tutorial onboarding.
2. Choose Independent Miner or Corporate Astronaut.
3. Acquire starter equipment and leave the station.
4. Mine nearby asteroids or salvage accessible wreckage.
5. Carry material in player or ship inventory.
6. Sell material at an exchange or use it for construction.
7. Establish a first outpost, base, vessel, or industrial foothold.
8. Improve mobility, storage, protection, and production.

Mid-game loop

Build specialised ships and machinery, establish reliable routes, refine materials, improve security, enter deeper regions, form crews or organisations, and compete for strategic resources and locations.

08 · SURVIVAL, OXYGEN & INVENTORY

Meaningful pressure, not a chemistry degree

Oxygen

Players require oxygen in vacuum, hazardous atmospheres, damaged ships, unpressurised bases, and hostile planetary environments. Each player wears a compact emergency oxygen backpack and a capable, streamlined helmet suitable for industrial work, travel, mining, and combat.

Design intent: Oxygen is an operational constraint, not a tedious spreadsheet. It should reward planning, airlocks, resupply, ship interiors, and recovery without demanding advanced atmospheric simulation for ordinary play.

Proposed oxygen rules

- Limited personal oxygen reserve in unsafe environments.
- Refills at stations, bases, airlocks, ships, refill points, and portable equipment.
- Escalating warnings and failure risk when supply runs low.
- Breaches, power loss, or damaged infrastructure can threaten oxygen availability.

Inventory philosophy

Layer	Purpose	Examples
Player inventory	Limited personal carry capacity. The player is not a walking warehouse.	Raptor, emergency supplies, oxygen, weapons, small mining yields, repair materials, components.
Ship inventory	Mobile logistics and role-defining cargo capacity.	Ore holds, containers, lockers, refined stores, ammunition, restricted storage.
Base / station storage	Industrial scale storage, production, exchange, and player organisation infrastructure.	Warehouses, refineries, fabrication stores, docking links, shared logistics.

09 · ECONOMY & RESOURCES

Materials need a reason to exist

Exchanges

Major stations contain exchanges where players buy and sell mined or salvaged material, parts, equipment, oxygen, supplies, and construction modules. Exchanges bridge the early mining loop to wider progression.

Independent economy

Independents retain freedom to choose contracts and player-to-player opportunities, but receive less favourable corporate exchange rates.

Corporate economy

Corporate Astronauts receive better rates and easier access to corporate infrastructure, but pay in obligations, quotas, and reduced autonomy.

Resource flow

Asteroid / Salvage / Planetary Site
-> Player or Ship Collection
-> Cargo Storage
-> Refining / Processing / Exchange
-> Fabrication / Construction / Trade / Repair

Initial resource set

PROPOSED Start with a compact, readable material set: common structural ore; conductive material for power and electronics; industrial alloy stock; fuel or energy material; a rarer advanced material deeper in the system; and recoverable salvaged components.

10 • MINING & ASTEROIDS

The Raptor and editable frontier matter

Raptor tool

The Raptor is the initial handheld mining tool. It removes material from mineable asteroid volumes, salvage targets, and potentially damaged structures or deposits. It should feel precise, industrial, and useful rather than magical.

Current prototype evidence

Player-view mining traces, real density carving, Marching Cubes remeshing, collision refresh, a generic voxel-mineable interface, and basic loose-fragment cleanup.

Future Raptor layer

Tool efficiency, resource yields, material behaviours, impact feedback, dust, sparks, beam effects, sound, range, precision, and specialised upgrades.

Asteroid spawners

Asteroid spawners populate designated regions with mineable content and should be server-authoritative. They should support conditions such as region type, distance from stations, planetary orbit, asteroid belt membership, resource tier, security level, Spacegate depth, player activity, and event-driven operations.

Asteroid belts around planets

Some planets should have belts, rings, debris fields, or orbital mining zones. These can provide dense common resources, navigation hazards, corporate claims, salvage, rare pockets, ambush opportunities, and spatially interesting mining routes.

Spawner rules: never spawn into player structures, ships, stations, or active safe areas; respect infrastructure distance; treat valuable locations as rarer and more dangerous; preserve actively mined or claimed objects; use controlled regeneration for abandoned regions.

Technical direction: Final asteroid visuals should use authored / scanned-style asteroid source meshes processed into editable voxel density assets. The current voxel sphere is a successful test platform, not the final aesthetic.

11 · BASES, SHIPS & SHIP EDITOR

Construction should be powerful without becoming punishment

Player bases

Players need outposts, industrial sites, dockyards, storage yards, refineries, defences, and staging areas. The first base is a major early milestone: the moment a player stops merely operating in Voraxia Prime and begins contributing to its infrastructure.

Base functions

Storage, refining, fabrication, docking, ship repair, oxygen resupply, power distribution, defence, processing, trade, permissions, and player-organisation infrastructure.

Modular ship editor

Voraxia needs a major in-game ship editor focused on modularity, clarity, and satisfying iteration. Players should build useful vessels quickly, then gradually gain access to deeper engineering and customisation.

Standard construction mode

Place hull modules, engines, storage, power systems, cargo bays, cockpits, airlocks, docking ports, interior modules, weapons, and defences. Give clear feedback on attachment, connectivity, function, balance, and missing requirements.

Advanced construction mode

PROPOSED Detailed layouts, internal systems, advanced power routing, specialised propulsion, automation, logic, complex industrial systems, higher efficiency, and expert trade-offs.

Potential ship roles

Starter shuttle, personal transport, miner, cargo hauler, salvage craft, scout, explorer, escort, combat vessel, mobile refinery, station tender, orbital builder, and planetary lander.

12 • PLANETARY EXPLORATION

From ship to surface and back again

Players can visit planets. Planetary areas add resource, exploration, construction, survival, and combat opportunities beyond open space.

Planetary content

Surface mining zones, corporate installations, ruins, salvage sites, hazardous environments, player outposts, landing zones, caves, ravines, deserts, ice fields, industrial debris, and planet-specific materials.

Planetary pressure

Gravity, terrain, atmosphere, weather, limited landing zones, surface PvP, fixed resource deposits, and structures that are visible, vulnerable, or defensible in different ways.

Ship-to-planet travel

PROPOSED Travel should feel like a coherent journey. Players use their ships between orbit, approach, landing zones, and surface locations. Streaming, atmospheric approach sequences, or carefully disguised boundaries are acceptable where needed. The priority is believable travel and useful gameplay, not technically expensive seamlessness for its own sake.

Scope principle: Full planet-scale seamless simulation is not mandatory for the first version. The player experience must preserve the feeling of leaving orbit, descending, landing, working, and returning to space.

13 • COMBAT & PVP

Conflict with consequence

PvP is a real part of Voraxia beyond protected station space. It should create tension, piracy, defence, territorial conflict, recovery, and cooperation, rather than becoming consequence-free disruption.

Character combat

Personal weapons and combat-capable tools used around asteroid sites, player bases, ships, planets, salvage operations, boarding events, defence, escape, and recovery.

Ship combat

Weapons and defensive systems tied to engineering choices: power, armour, mobility, layout, cargo load, crew, damage state, and tactical circumstance.

Defence and recovery

Players protect assets through armour, weapons, defensive modules, sensors, alarms, placement, crew cooperation, escort, corporate support, and player alliances. Damage should create decisions around repair, escape, salvage, cargo loss, resource depletion, crew safety, reputation, ownership, and recovery.

Open system: Exact death, respawn, insurance, loot, destruction, salvage, and recovery rules are still to be defined.

14 • MULTIPLAYER-FIRST PRINCIPLES

Shared universe truth belongs to the server

Voraxia is multiplayer-first. Major systems must be designed around authority, persistence, fairness, performance, interaction, recovery, and exploit resistance.

Server authority

- Player and ship inventory.
- Resource extraction and voxel mining edits.
- Asteroid state and spawner behaviour.
- Construction, ship, base, ownership, and permissions.
- Exchange transactions and trade.
- Weapon validation, combat damage, and PvP outcomes.
- Persistent world changes.

Client responsibility

- Responsive visual feedback.
- Local presentation, sound, interface, camera feedback, and effects.
- Prediction where appropriate.
- Never being trusted as the final source of permanent world truth.

“A feature is not complete when it works alone. It is complete when it works predictably, fairly, securely, and performantly in a shared universe.”

MULTIPLAYER-FIRST RULE

15 • CURRENT PROTOTYPE & FIRST VERTICAL SLICE

What is already proven

- Third-person character movement, sprinting, focus, and shoulder swapping.
- Custom pivot-based camera with smoothing, focus support, collision handling, dithering occlusion, and shake support.
- Voxel sphere generation through Marching Cubes.
- Real density-based asteroid carving, mesh remeshing, and collision refresh.
- Raptor-style mining trace from the player's view.
- Generic mineable interface and prototype detached-fragment cleanup.

First playable slice

World

One major starter station, one corporation, one nearby asteroid field, one small hostile frontier area, and a focused multiplayer test environment.

Player loop

Two origin paths, one Raptor tool, basic oxygen, player inventory, cargo inventory, exchange, first base loop, and first ship loop.

Resource loop

One basic asteroid resource, collection, transport, sale or construction use, and a meaningful reason to return to station or base.

Design proof

Players can leave safety, gather, risk loss, create a foothold, and improve their ability to reach further opportunities.

16 · SCOPE DISCIPLINE & OPEN QUESTIONS

Build layers, not a galaxy-shaped to-do list

Voraxia is ambitious. It needs staged development so strong foundational systems arrive before advanced simulation, huge content breadth, or technical spectacle.

Later layers

LATER LAYER Additional resource types, planets, deep Spacegate regions, advanced ship systems, broader asteroid belts, salvage, player organisations, territory systems, automation, industrial machinery, complex survival, large-scale base warfare, authored-asteroid voxelisation, chunked remeshing, and robust persistence.

Immediate design questions

1. What is the corporation called?
2. What starter equipment does each origin path receive?
3. What is the first material, first machine, and first meaningful base module?
4. What is the first buildable or ownable ship?
5. What does a player lose on death, and what can be recovered, insured, or looted?
6. How strong is corporate protection, and where does it apply?
7. What is the first planet and first Spacegate destination?
8. What are the first PvP rules around stations, bases, ships, and players?
9. How much ship-to-planet travel is streamed, seamless, or instance-based?
10. What player actions create permanent world change in the earliest multiplayer build?

Next recommended GDD pass: define the first ten hours in practical detail: starter loadouts for both paths, first resource, first exchange transaction, first base module, first ship goal, first hostile encounter, death / recovery rule, and first reason to enter a Spacegate.

VORAXIA: REBIRTH

**Reclaim the
dangerous frontier.
Build a life within it.
Defend what you
create.**

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Prepared as a living design document. Confirmed systems,
proposed mechanics, and scope layers are intended to be
revised as Voraxia develops.